BINOMIAL AND NORMAL DISTRIBUTIONS

INTRODUCTION OF PROBABILITY:

The word probability has two basic meanings; (i) ***a numerical measure of uncertainty*** and **(ii)** ***a measure of confidence in a given statement.***

What do we mean when we make the statement “Ali will probably win the tennis match”, “I have a fifty, fifty chances of getting an odd number when a die is tossed”, it is likely to rain today”, “I will probably pass the examination this year.” In all these statements, the uncertainty is expressed, but on the basis of past information or from the behavior of experiment, we have some degree of confidence in the validity of each statement.

The mathematical theory of probability for finite sample space provides a set of real numbers known as ***probabilities***, ranging from 0 to 1. ***It must be noted that the probability of the occurrence of sure sample point is equal to 1 and the probability of the occurrence of impossible sample point is 0.***

RANDOM EXPERIMENT:

Trial:

An individual performance for an experiment is called as trial.

e.g. tossing of a coin, rolling of a die etc.

Outcome:

The result of a trial is called as outcome.

e.g. When we toss a coin, the outcome may “Head” or “Tail”, when we roll a die the outcome may be 1, 2, 3, 4, 5 or 6 etc.

Random Experiment:

An experiment, in which results vary from trial to trial under similar conditions, is called as Random experiment.

Sample space:

A set consisting of all possible outcomes of an experiment is called as sample space and it is denoted by S.

For example; the sample space for a coin tossing experiment is;

$$S = \{H, T\}$$

Similarly when we roll a die then sample space is;

$$S = \{1, 2, 3, 4, 5, 6\} \text{ etc.}$$

Each possible outcome is the member of sample space *S* and is called as ***sample point*** in the sample space.

Event:

A subset of the sample space S is called an event.

For example; when we toss a coin, the sample space is S = {H, T} and the event that a head occurs is A= {H}

The events may be simple, compound, impossible, sure etc.

Simple Event:

An event which consists of only one sample point is called as simple event.

For example; when we toss two fair coins then the event that exactly two heads occur is A= {HH} is a simple event

Compound Event:

An event which consists of more than one sample points is called as compound event.

For example; when we toss two fair coins then the event that exactly one head occurs is A= {HT, TH} is a compound event

Impossible Event:

An event which contains no sample point is called an impossible event and it is denoted by ϕ .

Sure Event:

An event which contains all possible outcomes of an experiment is called as sure event and it is denoted by S.

Mutually Exclusive events:

Two events A and B of a single experiment are said to be mutually exclusive if and only if they cannot both occur at the same time or they have no common point i.e. $A \cap B = \phi$

For example; if A = {2, 4 , 6, 8, 10} and B = {1, 3, 5, 7, 9}

$$A \cap B = \{ \} = \phi$$

Therefore A and B are two mutually exclusive events.

Exhaustive events:

When the union of mutually exclusive events is equal to the entire sample space S, then these events are said to be collectively exhaustive i.e. $A \cup B = S$

For example; if S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {1, 3, 5, 7, 9} and B = {2, 4, 6, 8, 10}

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} = S$$

Therefore A and B are exhaustive events.

Equally Likely events:

Two events A and B are said to be equally likely, when they have equal chances of occurrence.

For example; when we toss a coin, then Head is as likely to occur as Tail, therefore Head and Tail are equally likely events.

DEFINITION OF PROBABILITY:

The Classical Definition of Probability:

If a random experiment can give "n" mutually exclusive and equally likely outcomes and if "m" of which are favorable to the occurrence of a certain event A, then the probability of the event A, denoted by P (A) is given as;

$$P(A) = \frac{m}{n} = \frac{\text{Number of favourable outcomes}}{\text{Number of all possible outcomes}}$$

Properties for the Probability of an event A:

(i) $0 \le P(A) \le 1$ **(ii)** Total probability = P (S) = 1

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Some Information about Playing Cards:

The face values of 13 cards for each suit are as under;

RED CARDS		BLACK CARDS	
Heart	**Diamond**	**Spades**	**Club**
A	A	A	A
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9
10	10	10	10
J	J	J	J
Q	Q	Q	Q
K	K	K	K
13	**13**	**13**	**13**

Therefore;

Number of Red cards = 26
Number of black cards = 26
Number of Aces = 4

Pictured Cards
Number of Jack Cards = 4
Number of Queen Cards = 4
Number of King Cards = 4

So Number of Pictured cards = 4 + 4 + 4 = 12
Number of face cards = 12 or 16

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Example# 1: **(a)** Find the probability of getting a head when toss a fair coin.
(b) A die is rolled; find the probability of getting a 3.
(c) A die is rolled; find the probability of getting an even number.

***Solution*:**

(a) The sample space for a coin is;

S = {H, T}

$$P(H) = \frac{1}{2}$$

(b) The sample space for a die is;

S = {1, 2, 3, 4, 5, 6}

$$P(getting\,3) = \frac{1}{6}$$

(c) The sample space for a die is;

S = {1, 2, 3, 4, 5, 6}

$$P(an\ even\ number) = \frac{3}{6} = \frac{1}{2}$$

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Example# 2: Two coins are tossed, find the probability of getting **(i)** only one head **(ii)** two heads **(iii)** two tails.

Solution: The sample space for two coins is

S = {HH, HT, TH, TT}

(a) $P(1\ Head) = \frac{2}{4} = \frac{1}{2}$

(b) $P(2\ Heads) = \frac{1}{4}$

(c) $P(2\ Tails) = \frac{1}{4}$

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Example# 3: Two fair dice are rolled together, find the probability that **(i)** sum of dots is equal to 5 **(ii)** sum of dots is even **(iii)** sum of dots is greater than 8 **(iv)** sum of dots is odd **(v)** product is between 4 and 12 (both inclusive) **(vi)** same number appears on both dice.

Solution: The sample space for pair of dice is;

S = **{**(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2 , 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3 , 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4 , 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5 , 6), (6 , 1), (6 ,2), (6 , 3), (6, 4), (6, 5), (6 , 6) **}**

(i) Let A = Sum of dots is 5

A = **{**(1, 4), (2, 3), (3, 2), (4, 1)**}**

$$P(A) = \frac{4}{36} = \frac{1}{9}$$

(ii) Let B = Sum of dots is even

B = **{**(1, 1), (1, 3), (1, 5), (2, 2), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (4, 6), (5, 1), (5, 3), (5, 5), (6, 2), (6, 4), (6, 6)**}**

$$P(B) = \frac{18}{36} = \frac{1}{2}$$

(iii) Let C = Sum of dots is greater than 8

C = **{**(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)**}**

$$P(C) = \frac{10}{36} = \frac{5}{18}$$

(iv) Let D = Sum of dot is odd

D = **{**(1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (3, 6), (4, 1), (4, 3), (4, 5), (5, 2), (5, 4), (5, 6), (6, 1), (6, 3), (6, 5)**}**

$$P(D) = \frac{18}{36} = \frac{1}{2}$$

(v) Let E = Product is between 4 and 12 (both inclusive)

E = **{**(1, 4), (1, 5), (1, 6), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (6, 1), (6, 2)**}**

$$P(C)=\frac{18}{36}=\frac{1}{2}$$

(vi) Let F = Same number appears on both dice.

F = **{**(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (5, 2), (6, 6)**}**

$$P(F)=\frac{6}{36}=\frac{1}{6}$$

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Note: In probability we use

OR = + **AND = x**

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Example# 4: A pair of dice is thrown. Find the probability of getting the sum of dots on the upper sides of the dice is **(i)** 4 or 7 **(ii)** 9 **(iii)** 11 **(iv)** even or less than 7.

Solution: The sample space for pair of dice is;

S = **{**(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3 , 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4 , 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6 , 1), (6 ,2), (6 , 3), (6, 4), (6, 5), (6 , 6) **}**

(i) Let A = Sum of dots is 4

Let B = Sum of dots is 7

A = **{**(1, 3), (2, 2), (3, 1)**}**

B = **{**(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)**}**

$$P(A\ or\ B)=P(A)+P(B)$$

$$P(A\ or\ B)=\frac{3}{36}+\frac{6}{36}=\frac{9}{36}=\frac{1}{4}$$

(ii) Let C = Sum of dot is eleven

C = **{**(5, 6), (6, 5)**}**

$$P(C)=\frac{2}{36}=\frac{1}{18}$$

(iii) Let D = Sum of dots is even

Let E = Sum is less than 7

D = **{**(1, 1), (1, 3), (1, 5), (2, 2), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (4, 6), (5, 1), (5, 3), (5, 5), (6, 2), (6, 4), (6, 6)**}**

E = **{**(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (5, 1)**}**

$$P(D\ or\ E)=P(D)+P(E)$$

D.Y.S.

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Example# 5: A jar contains 3 red marbles, 7 green marbles and 10 white marbles. If a marble is drawn from the jar at random, what is the probability that this marble is white?

Solution:

Red		Green		White		Total
3	+	7	+	10	=	20

P (White marble) = $\frac{10}{20} = \frac{1}{2}$

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Example# 6: A fair coin is tossed three times. What is the probability that at least one head occurs?

Solution: The sample space for a three dice is;

S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

Let A = At least one head appears

A = {HHH, HHT, HTH, HTT, THH, THT, TTH}

P (A) = $\frac{7}{8}$

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Example# 7: If a card is drawn from a pack of 52 playing cards, find the probability that it is a **(i)** red card **(ii)** card number is 10 **(iii)** diamond cards.

Solution:

(i) $P(\text{Red}\ Card) = \frac{26}{52} = \frac{1}{2}$

(ii) $P(Card\ \ number\ 10) = \frac{4}{52} = \frac{1}{13}$

(iii) $P(Diamond\ \ Card) = \frac{13}{52} = \frac{1}{4}$

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1. BINOMIAL PROBABILITY DISTRIBUTION:

Bernoulli Trials:

Repeated independent trials in which there are only two possible complementary outcomes denoted by S for success and F for failure with $P(S) = p$ and $P(F) = 1 - p = q$ (*i.e.* probability of success remains same for all trials) are called Bernoulli trials.

Binomial Experiment:

An experiment consisting of "*n*" Bernoulli trials is called as Binomial experiment.

OR

An experiment of "*n*" independent trials, in which the outcome can always be classified as either a success or a failure and in which probability of success remains constant from trial to trial is called a Binomial experiment.

Properties of Binomial Experiment:

(i) The outcome of each trial may be classified into one of two categories: success and failure.

(ii) The successive trials are all independent.

(iii) The probability of success remains constant from trial to trial.

(iv) The experiment is repeated a fixed number of times say "*n*".

Binomial Random Variable:

The random variable X which denotes the number of successes in a Binomial experiment is called as Binomial random variable.

Binomial Probability Function:

If p denotes the probability of success and $q = 1 - p$ denotes the probability of failure, then the probability distribution for exactly x success in n trials is

$$P(X = x) = b(x; n, p) = \binom{n}{x} p^x q^{n-x}; \qquad x = 0, 1, \ldots\ldots, n$$

called the Binomial Probability Distribution.

Where p = Probability of success.

$q = 1 - p$ = Probability of failure.

n = Number of trials

x = Number of successes

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Parameters of Binomial Probability distribution:

Binomial Probability Distribution has two parameters as n *and* p.

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Note: As we know that

Total probability = 1

$$q + p = 1$$

$$\Rightarrow \quad p = 1 - q$$

$$and \quad q = 1 - p$$

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Note: The desired probability always stands for success.

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Note: (Compliment Law)

(i) $P(X \geq x) = 1 - P(X < x)$

(ii) $P(X > x) = 1 - P(X \leq x)$

(iii) $P(X \leq x) = 1 - P(X > x)$

(iv) $P(X < x) = 1 - P(X \geq x)$

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Note:

(i) At least $\geq$ **(ii)** At most $\leq$ **(iii)** fewer than $<$ **(iv)** Not more than $\leq$

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<u>Example# 8</u>: Suppose you independently flip a coin 4 times and the outcome of each toss can be either head or tails. What is the probability of obtaining exactly 2 tails?

***Solution*:** Given $n=4,\qquad p=P(tail)=\frac{1}{2}$

then $q=1-p=1-\frac{1}{2}=\frac{1}{2}$

As $P(X=x)=\binom{n}{x}p^x q^{n-x}=\binom{4}{x}\left(\frac{1}{2}\right)^x\left(\frac{1}{2}\right)^{4-x}\qquad x=0,1,2,3,4$

$$P(X=2)=\binom{4}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{4-2}$$

$$P(X=2)=6\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)^2=6\left(\frac{1}{4}\right)\left(\frac{1}{4}\right)=\frac{6}{16}$$

$$P(X=2)=\frac{3}{8}$$

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Example# 9: Let X have a binomial distribution with $n = 3$ and $p = 0.4$. Find $P\left(X=\frac{3}{2}\right)$, $P(X=2)$, $P(X\le 2)$, $P(X=-2)$ *and* $P(X\ge 2)$

***Solution*:** Given $n=3,\qquad p=0.4$

then $q=1-p=1-0.4=0.6$

As $P(X=x)=\binom{n}{x}p^x q^{n-x}=\binom{3}{x}(0.4)^x(0.6)^{3-x};\qquad x=0,1,2,3$

(i) $P(X=\frac{3}{2})=0$

Because there is no probability for fraction in Binomial Probability distribution

(ii) $P(X=2)=\binom{3}{2}(0.4)^2(0.6)^{3-2}=(3)(0.16)(0.6)=0.288$

(iii) $P(X\le 2)=P(X=0)+P(X=1)+P(X=2)$

$$P(X\le 2)=\binom{3}{0}(0.4)^0(0.6)^{3-0}+\binom{3}{1}(0.4)^1(0.6)^{3-1}+\binom{3}{2}(0.4)^2(0.6)^{3-2}$$

$$P(X\le 2)=(1)(1)(0.216)+(3)(0.4)(0.36)+(3)(0.16)(0.6)$$

$$P(X\le 2)=0.216+0.432+0.288=0.936$$

(iv) $P(X=-2)=0$

Because there is no probability for Negative numbers in Binomial Probability distribution

(v) $P(X\ge 2)=P(X=2)+P(X=3)$

$$P(X \geq 2) = \binom{3}{2}(0.4)^2(0.6)^{3-2} + \binom{3}{3}(0.4)^3(0.6)^{3-3}$$

$$P(X \leq 2) = (3)(0.16)(0.6) + (1)(0.064)(1)$$

$$P(X \leq 2) = 0.288 + 0.064 = 0.352$$

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Example# 10: A die is rolled five times and a 5 or 6 is considered a success. Find the probability of **(i)** no success **(ii)** at least 2 success

***Solution*:** Given $n = 5$

$$p = P(5 \; or \; 6) = P(5) + P(6)$$

$$p = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

then $$q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

Let X = Number of getting 5 or 6 on a die

As $$P(X = x) = \binom{n}{x} p^x q^{n-x} = \binom{5}{x}\left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{5-x} ; \qquad x = 0, 1, 2, 3, 4, 5$$

(i) $$P(X = 0) = \binom{5}{0}\left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{5-0} = (1)(1)\left(\frac{32}{243}\right) = \frac{32}{243}$$

(ii) $$P(X \geq 2) = 1 - P(X < 2)$$

$$P(X \geq 2) = 1 - \left[P(X = 0) + P(X = 1)\right]$$

$$P(X \geq 2) = 1 - \left[\binom{5}{0}\left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{5-0} + \binom{5}{1}\left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^{5-1}\right]$$

$$P(X \geq 2) = 1 - \left[(1)(1)\left(\frac{32}{243}\right) + (5)\left(\frac{1}{3}\right)\left(\frac{16}{81}\right)\right]$$

$$P(X \geq 2) = 1 - \left[\frac{32}{243} + \frac{80}{243}\right] = 1 - \frac{112}{243} = \frac{131}{243} = 0.539$$

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Example# 11: Suppose you independently throw a dart 10 times. Each time you throw a dart, the probability of hitting the target is $\frac{3}{4}$. What is the probability of hitting the target less than 5 times?

Example# 12: A clothing store has determined that 30% of the people who enter the store will make a purchase. Eight people enter the store during a one-hour period. Find the probability that (a) exactly four people will make a purchase, and (b) at least one person will make a purchase.

Example# 13: Seven out of every 100 people in the United States have Type O negative blood. If 6 people are chosen at random, what is the probability that exactly 2 of them have Type O negative blood?

D.Y.S.

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Example# 14: Find the probability of getting **(i)** exactly 4 heads and **(ii)** not more than 4 heads when 6 coins are tossed.

Solution: Given $n=6,\qquad p=P(Head)=\dfrac{1}{2}$

then $q=1-p=1-\dfrac{1}{2}=\dfrac{1}{2}$

Let X = Number of heads on a die.

As $P(X=x)=\binom{n}{x}p^x q^{n-x}=\binom{6}{x}\left(\dfrac{1}{2}\right)^x\left(\dfrac{1}{2}\right)^{6-x};\qquad x=0,1,2,3,4,5,6$

(i) $P(X=4)=\binom{6}{4}\left(\dfrac{1}{2}\right)^4\left(\dfrac{1}{2}\right)^{6-4}=(15)\left(\dfrac{1}{16}\right)\left(\dfrac{1}{4}\right)=\dfrac{15}{64}$

(ii) $P(X\le 4)=1-P(X>4)$

$$P(X\le 4)=1-\left[P(X=5)+P(X=6)\right]$$

$$P(X\le 4)=1-\left[\binom{6}{5}\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^{6-5}+\binom{6}{6}\left(\frac{1}{2}\right)^6\left(\frac{1}{2}\right)^{6-6}\right]$$

$$P(X\le 4)=1-\left[(6)\left(\frac{1}{32}\right)\left(\frac{1}{2}\right)+(1)\left(\frac{1}{64}\right)(1)\right]$$

$$P(X\ge 2)=1-\left[\frac{6}{64}+\frac{1}{64}\right]=1-\frac{7}{64}=\frac{57}{64}$$

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Example# 15: If 60% of the voters in a large district prefer candidate *A*, what is the probability that in a sample of 12 voters exactly 7 will prefer *A*?

Solution: Given $n=12,\qquad p=60\%=0.60$

then $q=1-p=1-0.60=0.40$

Let X = Number of voters who prefer candidate A

As $P(X=x)=\binom{n}{x}p^x q^{n-x}=\binom{12}{x}(0.60)^x(0.40)^{12-x};\qquad x=0,1,2,3,\ldots\ldots,12$

$P(X=7)=?$ **D.Y.S.**

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Example# 16: The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly five of the next 7 patients having this operation survive?

Solution: Given $n=7,\qquad p=0.9$

then $q=1-p=1-0.9=0.1$

Let X = Number of recovered patients from heart operation

As $P(X=x)=\binom{n}{x}p^x q^{n-x}=\binom{7}{x}(0.9)^x(0.1)^{7-x};\qquad x=0,1,2,3,4,5,6,7$

$P(X=5)=?$ **D.Y.S.**

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Example# 17: One in every 6 coffee drinkers at a restaurant prefers coffee with cream, sugar, or both. If 4 coffee drinkers are chosen at random, what is the probability that at least 1 coffee drinker prefers coffee with cream, sugar, or both?

Example# 18: Eighty percent of the songs played at a radio station are rock songs. If 10 songs are chosen at random from the station's play list, what is the probability that at least 9 of the songs will be rock songs? **D.Y.S.**

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Example# 19: The incidence of occupational disease in an industry is such that the workmen have a 20% chance of suffering from it. What is the probability that out of 6 workmen **(i)** not more than 2 and **(ii)** 4 or more will catch the disease?

Solution: Given $n = 6,\quad p = 20\% = 0.20$

then $q = 1 - p = 1 - 0.20 = 0.80$

Let X = Number of workmen suffering from occupational disease.

As $P(X = x) = \binom{n}{x} p^x q^{n-x} = \binom{6}{x} (0.20)^x (0.80)^{6-x}; \qquad x = 0, 1, 2, 3, 4, 5, 6$

(i) $P(X \le 2) = ?$

(ii) $P(X \ge 4) = ?$ **D.Y.S.**

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<u>**Binomial Frequency Distribution**</u>:

If the binomial probability distribution is multiplied by N, the number of experiments, the resulting distribution is known as Binomial frequency distribution. Thus the expected frequency of x successes in "N" experiment is $N.\binom{n}{x} p^x q^{n-x}$

i.e. $$f_e = N.\binom{n}{x} p^x q^{n-x} = N.(q+p)^n$$

where f_e = Expected or Theoretical frequency

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Example# 20: Five dice are tossed 96 times. Find the expected frequencies when throwing of a 5 or a 6 is called as a success.

Solution: Given $n = 5,\quad N = 96$

$$p = P(4,5 \text{ or } 6) = P(4) + P(5) + P(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

then $q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$

Let X = Number of getting of 4, 5 or 6 on a die.

As $P(X = x) = \binom{n}{x} p^x q^{n-x} = \binom{5}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$

x	$P(X=x)=\binom{5}{x}\left(\frac{1}{2}\right)^{x}\left(\frac{1}{2}\right)^{5-x}$	$f_e = N.P(X=x)$
0	$\binom{5}{0}\left(\frac{1}{2}\right)^{0}\left(\frac{1}{2}\right)^{5-0}=\frac{1}{32}$	$96\times\frac{1}{32}=3$
1	$\binom{5}{1}\left(\frac{1}{2}\right)^{1}\left(\frac{1}{2}\right)^{5-1}=\frac{5}{32}$	$96\times\frac{5}{32}=15$
2	$\binom{5}{2}\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{5-2}=\frac{10}{32}$	$96\times\frac{10}{32}=30$
3	$\binom{5}{3}\left(\frac{1}{2}\right)^{3}\left(\frac{1}{2}\right)^{5-3}=\frac{10}{32}$	$96\times\frac{10}{32}=30$
4	$\binom{5}{4}\left(\frac{1}{2}\right)^{4}\left(\frac{1}{2}\right)^{5-4}=\frac{5}{32}$	$96\times\frac{5}{32}=15$
5	$\binom{5}{5}\left(\frac{1}{2}\right)^{5}\left(\frac{1}{2}\right)^{5-5}=\frac{1}{32}$	$96\times\frac{1}{32}=3$
Total	**1.0000**	$\sum f_e = 96$

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<u>Note</u>:

In Binomial Probability distribution

$$Mean = \mu = np$$
$$Variance = \sigma^2 = npq$$
$$S.D. = \sigma = \sqrt{npq}$$

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<u>Note</u>: In Binomial Probability distribution

$$\mu > \sigma^2$$

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Example# 21: A *r.v.* X is binomially distributed with mean 3 and variance 2. Find n and p.

Solution: Given

$$\mu = np = 3 \ldots\ldots\ldots\ldots\ldots\ldots(i)$$
$$\sigma^2 = npq = 2 \ldots\ldots\ldots\ldots\ldots\ldots(ii)$$

Putting $np = 3$ in equation (ii)

$$3q = 2 \quad \Rightarrow \quad q = \frac{2}{3}$$

And $\quad p = 1 - q = 1 - \frac{2}{3} = \frac{1}{3}$

Putting $p = \frac{1}{3}$ in equation (i)

$$n\left(\frac{1}{3}\right) = 3 \qquad \Rightarrow \qquad n = 3 \times 3 = 9$$

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Example# 22: Find the mean, variance and the standard deviation of a binomial distribution with 50 trials, for which the probability of success is 0.1?

***Solution*:** Given $n = 50, \qquad p = 0.1$

then $q = 1 - p = 1 - 0.1 = 0.90$

As we know that

$$\mu = np = (50)(0.1) = 5$$

$$\sigma^2 = npq = 50(0.1)(0.9) = 4.5$$

And $\sigma = \sqrt{npq} = \sqrt{50(0.1)(0.9)} = \sqrt{4.5} = 2.12$

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Example# 23: Find the standard deviation of a binomial distribution with 8 trials, for which the probability of success is 0.3.

Example# 24: In a doctor's office, 80% of the patients are adults. If 15 patients are scheduled for an appointment on a given day, what is the expected number of adults? What is the standard deviation?

D.Y.S.

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<u>Note</u>:

The shape of the Binomial probability distribution depends upon its parameters "*n*" and "*p*" as

(i) If $p = q = \frac{1}{2}$, then distribution is symmetrical.

(ii) If $p < q \;\; or \;\; p < \frac{1}{2}$, then distribution is positively skewed

(iii) If $p > q \;\; or \;\; p > \frac{1}{2}$, then distribution is negatively skewed.

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2. <u>NORMAL DISTRIBUTION</u>:

If "*n*" (no. of trials) in a binomial probability distribution becomes very large and "*p*" (probability of success) does not become very small, then the binomial probability distribution approaches a continuous distribution known as "Normal Distribution".

OR

The limiting form of the binomial probability distribution is known as normal distribution

i.e $\lim_{n \to \infty} b(x; n, p) = Normal\ Distribution$

The Normal distribution is also called the Gaussian distribution in honor of the great German mathematician Carl F. Gauss (1777 – 1855), who derived its equation mathematically as the probability distribution of the errors of measurement.

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EQUATION OF NORMAL DISTRIBUTION OR THE PROBABILITY DENSITY FUNCTION (*p.d.f*) OF THE NORMAL DISTRIBUTION:

The *p.d.f.* of the Normal Distribution is;

$$y = f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2}; \qquad -\infty \le X \le \infty$$

Where

$y =$ The values marked along Y-axis (Ordinate)

$x =$ The values marked along X-axis (Abscissa)

$\mu =$ Mean of the normal distribution

$\sigma =$ Standard deviation of the normal distribution

$\pi =$ Constant $\left(\frac{22}{7}\right)$

$e =$ Constant $(2.71828182.....)$

……………………………………………………

Parameters of Normal distribution:

Normal distribution has two parameters μ *and* σ^2 where μ is the location parameter and σ^2 is the shape parameter.

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Note:

$X \sim N(\mu, \sigma^2)$ is read as "X is distributed Normally with mean μ and variance σ^2".

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Standardized Variable:

The variable $Z = \frac{X-\mu}{\sigma}$ which measures the deviation from mean in standard deviation units is called the standardized variable or standard normal variable. ***It is important to note that the standardized variable always has mean zero and variance one.***

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Standardized Normal Distribution:

The normal probability distribution of Z which has zero mean and unit variance, is called the *Standardized Normal distribution or Unit Normal distribution* and is denoted by $N(0, 1)$. The *p.d.f.* for *Standard Normal* distribution denoted by $\phi(z)$ is given as;

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; \qquad -\infty \le z \le \infty$$

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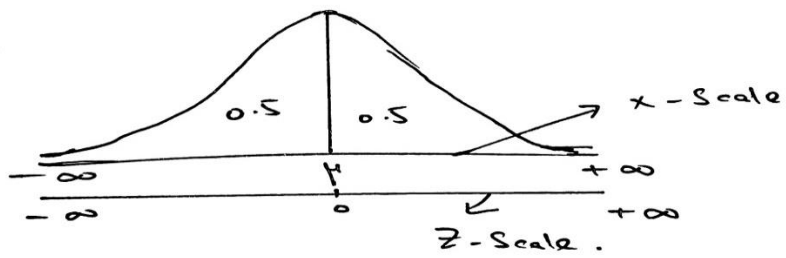
Shape of the Normal distribution:



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PROPERTIES OF NORMAL DISTRIBUTION:

The Normal distribution has following important properties:

(i) The distribution is bell-shaped, unimodel and symmetrical *i.e.* mean=median=mode

(ii) The total area under the normal curve is unity or one.

(iii) In a normal distribution points of inflection are equidistant from mean which lies at $\mu-\sigma$ and $\mu+\sigma$.

(iv) In a normal distribution $M.D = 0.7979\sigma = \frac{4}{5}\sigma$.

(v) In a normal distribution $Q.D = 0.6745\sigma$.

(vi) In a normal distribution all odd ordered moments about mean are zero.
i.e. $\mu_1 = \mu_3 = \mu_5 = ... = 0$. Also $\beta_1 = 0$.

(vii) In a normal distribution $\mu_2 = \sigma^2$, $\mu_4 = 3(\sigma^2)^2$ and $\beta_2 = 3$.

(viii) In a normal distribution $Q_1 = \mu - 0.6745\sigma$ and $Q_3 = \mu + 0.6745\sigma$.

(ix) In a normal distribution maximum ordinate lies at $X = \mu$.

(x) In a normal distribution **(a)** $\mu \pm \sigma$ contains 68.27% of the total area **(b)** $\mu \pm 2\sigma$ contains 95.45% of the total area **(c)** $\mu \pm 3\sigma$ contains 99.73% of the total area.

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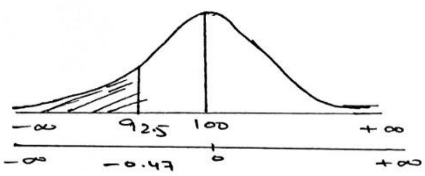
Example# 25: Let the random variable X is normally distributed with mean 100 and variance 256, then find the following probabilities;

(a) $P(X \le 92.5)$ **(b)** $P(X \le 107.5)$ **(c)** $P(X \ge 124)$ **(d)** $P(112 \le X \le 128.5)$

(e) $P(91 \le X \le 127)$ **(f)** $P(X \ge 76)$

***Solution*:** Given; $\mu = 100$, $\sigma^2 = 256 \Rightarrow \sigma = 16$

(a) $P(X \le 92.5) = ?$



$$P(X \le 92.5) = P\left(\frac{X-\mu}{\sigma} \le \frac{92.5-100}{16}\right)$$

$$P(X \le 92.5) = P(Z \le -0.47)$$

$$P(X \le 92.5) = \phi(-0.47)$$

$$P(X \le 92.5) = 0.3192$$

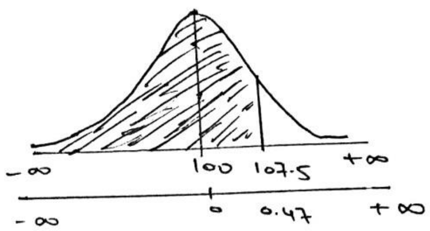
(b) $P(X \le 107.5) = ?$

$$P(X \le 107.5) = P\left(\frac{X-\mu}{\sigma} \le \frac{107.5-100}{16}\right)$$

$$P(X \le 107.5) = P(Z \le 0.47)$$

$$P(X \le 107.5) = \phi(0.47)$$

$$P(X \le 107.5) = 0.6808$$



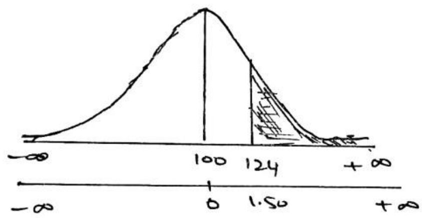
(c) $P(X \ge 124) = ?$

$$P(X \ge 124) = P\left(\frac{X-\mu}{\sigma} \ge \frac{124-100}{16}\right)$$

$$P(X \ge 124) = P(Z \ge 1.50)$$

$$P(X \ge 124) = 1-\phi(1.50)$$

$$P(X \ge 124) = 1-0.9332 = 0.0668$$



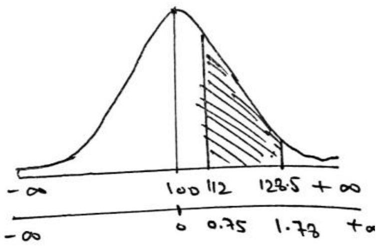
(d) $P(112 \le X \le 128.5) = ?$

$$P(112 \le X \le 128.5) = P\left(\frac{112-100}{16} \le \frac{X-\mu}{\sigma} \le \frac{128.5-100}{16}\right)$$

$$P(112 \le X \le 128.5) = P(0.75 \le Z \le 1.78)$$

$$P(112 \le X \le 128.5) = \phi(1.78) - \phi(0.75)$$

$$P(112 \le X \le 128.5) = 0.9625 - 0.7734 = 0.1891$$



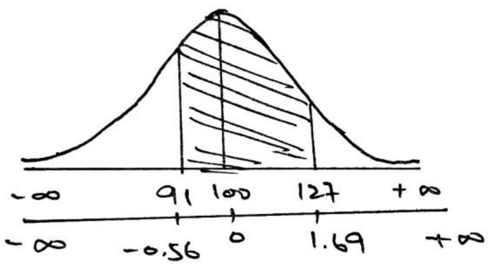
(e) $P(91 \le X \le 127) = ?$

$$P(91 \le X \le 127) = P\left(\frac{91-100}{16} \le \frac{X-\mu}{\sigma} \le \frac{127-100}{16}\right)$$

$$P(91 \le X \le 127) = P(-0.56 \le Z \le 1.69)$$

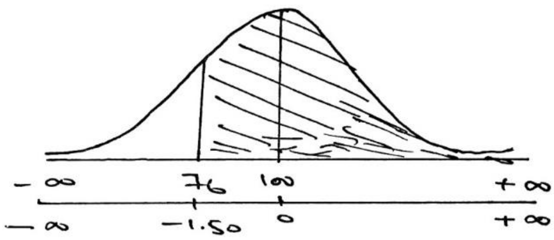
$$P(91 \le X \le 127) = \phi(1.69) - \phi(-0.56)$$

$$P(91 \le X \le 127) = 0.9545 - 0.2877 = 0.6$$



(f) $P(X \ge 76) = ?$

$$P(X \ge 76) = P\left(\frac{X-\mu}{\sigma} \ge \frac{76-100}{16}\right)$$



$$P(X \geq 76) = P(Z \geq -1.50)$$

$$P(X \geq 76) = 1 - \phi(-1.50)$$

$$P(X \geq 76) = 1 - 0.0606 = 0.9394$$

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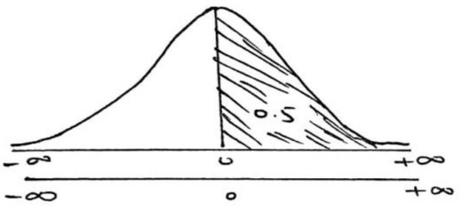
Example# 26: If X is normally distributed with zero mean and $\sigma = 0.6$, find $P(X > 0)$ and $P(0.2 < X < 1.8)$

Solution:

Given $\mu = 0$, $\sigma = 0.6$

(i) $P(X > 0) = 0.5$



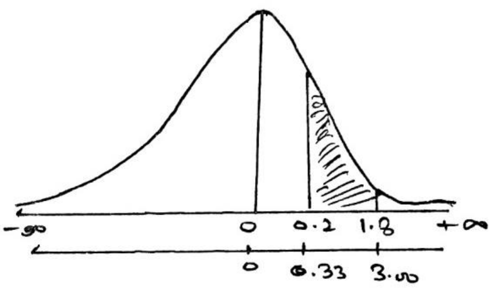
(ii) $P(0.2 < X > 1.8) = ?$

$$P(0.2 \leq X \leq 1.8) = P\left(\frac{0.2-0}{0.6} \leq \frac{X-\mu}{\sigma} \leq \frac{1.8-0}{0.6}\right)$$

$$P(0.2 < X < 1.8) = P(0.33 \leq Z \leq 3.00)$$

$$P(0.2 < X < 1.8) = \phi(3.00) - \phi(0.33)$$

$$P(0.2 < X < 1.8) = 0.9987 - 0.6293 = 0.3694$$



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Example# 27: Scores on a certain nation-wide college entrance examination follow a normal distribution with a mean of 500 and a standard deviation of 100. Find the probability that a student will score **(i)** over 650 **(ii)** less than 250 **(iii)** between 325 and 675.

Example# 28: A radar unit is used to measure speeds of cars on a motorway. The speeds are normally distributed with a mean of 90 km/hr and a standard deviation of 10 km/hr. What is the probability that a car picked at random is travelling at more than 100 km/hr?

Example# 29: The length of life of an instrument produced by a machine has a normal distribution with a mean of 12 months and standard deviation of 2 months. Find the probability that an instrument produced by this machine will last (a) less than 7 months.(b) between 7 and 12 months.

Example# 30: A large group of students took a test in Physics and the final grades have a mean of 70 and a standard deviation of 10. If we can approximate the distribution of these grades by a normal distribution, what percent of the students (a) scored higher than 80 (b) should pass the test (grades≥60) (c) should fail the test (grades<60)?

Example# 31: Scores on a standard IQ Test for people aged 20 to 34 are normally distributed with mean 110 and standard deviation 25.

(a) What percent of young adults below 75 on this IQ Test?

(b) What is the probability that a young adult will score above 180 on this IQ Test?

(c) What is the probability that a young adult will score between 85 and 160 on this IQ Test?

D.Y.S.

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Example# 32: Given that the heights of college boys is normally with mean 62 inches and standard deviation 4 inches, and the minimum height required for joining the police force is 64 inches. Find the percentage of boys who would be rejected on account of their height?

***Solution*:**

Given $\mu = 62''$, $\sigma = 4''$

Let X = Height required for joining *Police force*

$P(X \le 64'') = ?$

$$P(X \le 64'') = P\left(\frac{X-\mu}{\sigma} \le \frac{64-62}{4}\right)$$

$$P(X \le 64'') = P(Z \le 0.50)$$

$$P(X \le 64'') = \phi(0.50)$$

$$P(X \le 64'') = 0.6915$$

Hence required percentage of boys who would be rejected due to their height = 0.6915 x 100

Hence required percentage of boys who would be rejected due to their height = 69.15%

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NORMAL FREQUENCY DISTRIBUTION:

If we multiplied the normal distribution by "N" the number of sets of all possible experiments, we get the Normal frequency distribution as;

$$f_e = N.\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; \qquad -\infty \le X \le \infty$$

Where f_e = expected or theoretical frequency

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Example# 33: Suppose that the weights of 2000 male students are normally distributed with mean 155 pounds and standard deviation 20 pounds. Find the number of students with weight **(i)** less than or equal to 100 pounds **(ii)** between 120 and 130 pounds **(iii)** greater than or equal to 200 pounds.

***Solution*:** Given $N = 2000,$ $\mu = 155\ lbs$ *and* $\sigma = 20\ lbs$

Let X = Weight of students in pounds.

(i) $P(X \le 100) = ?$

$$P(X \le 100) = P\left(\frac{X-\mu}{\sigma} \le \frac{100-155}{20}\right)$$

$$P(X \le 100) = P(Z \le -2.75)$$

$$P(X \le 100) = \phi(-2.75)$$

$$P(X \le 100) = 0.0030$$

Hence number of students whose weight is less than or equal to 100 pounds = 0.0030 x N

Hence number of students whose weight is less than or equal to 100 pounds = 0.0030 x 2000

Hence number of students whose weight is less than or equal to 100 pounds = 6

(ii) $P(120 \le X \le 130) = ?$

$$P(120 \le X \le 130) = P\left(\frac{120-155}{20} \le \frac{X-\mu}{\sigma} \le \frac{130-155}{20}\right)$$

$$P(120 \le X \le 130) = P(-1.75 \le Z \le -1.25)$$

$$P(120 \le X \le 130) = \phi(-1.25) - \phi(-1.75)$$

$$P(120 \le X \le 130) = 0.1056 - 0.0401 = 0.0655$$

Hence number of students whose weight is between 120 and 130 pounds = 0.0655 x 2000
Hence number of students whose weight is between 120 and 130 pounds = 131

(iii) $P(X \ge 200) = ?$

$$P(X \ge 200) = P\left(\frac{X-\mu}{\sigma} \ge \frac{200-155}{20}\right)$$

$$P(X \ge 200) = P(Z \ge 2.25)$$

$$P(X \ge 200) = 1 - \phi(2.25)$$

$$P(X \ge 200) = 1 - 0.9878 = 0.0122$$

Hence number of students whose weight is greater than or equal to 200 pounds = 0.0122 x 2000
Hence number of students whose weight are greater than or equal to 200 pounds = 24

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Example# 34: The amount of mustard dispensed from a machine at The Hotdog Emporium is normally distributed with a mean of 0.9 ounce and a standard deviation of 0.1 ounce. If the machine is used 500 times, approximately how many times will it be expected to dispense 1 or more ounces of mustard?

Example# 35: Professor Halen has 184 students in his college mathematics lecture class. The scores on the midterm exam are normally distributed with a mean of 72.3 and a standard deviation of 8.9. How many students in the class can be expected to receive a score between 82 and 90? Express answer to the nearest student. **D.Y.S.**

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<u>USE OF INVERSE STANDARD NORMAL CUMULATIVE</u>:

Example# 36: If $X \sim N(70, 25)$, find;

(i) A point that has 87.9% of the distribution below it.
(ii) A point that has 81.7% of the distribution above it.
(iii) Two such points between which the central 70% of the distribution lies.

Solution: Given $\mu = 70$ *and* $\sigma^2 = 25 \Rightarrow \sigma = 5$

(i) Let "*a*" be the required point, then

$$P(X < a) = 87.9\%$$

$$P(X < a) = 0.879$$

$$P\left(\frac{X-\mu}{\sigma} < \frac{a-70}{5}\right) = 0.879$$

$$P\left(Z < \frac{a-70}{5}\right) = 0.879$$

$$\phi\left(\frac{a-70}{5}\right)=0.879$$

$$\frac{a-70}{5}=\phi^{-1}(0.879)$$

$$\frac{a-70}{5}=1.1700 \quad \Rightarrow \quad a-70=5(1.1700)$$

$$a-70=5.85 \quad \Rightarrow \quad a=70+5.85$$

$$a=75.85$$

(ii) Let "*a*" be the required point, then

$$P(X>a)=81.7\%$$

$$P(X>a)=0.817$$

$$P\left(\frac{X-\mu}{\sigma}>\frac{a-70}{5}\right)=0.817$$

$$P\left(Z>\frac{a-70}{5}\right)=0.817$$

$$1-\phi\left(\frac{a-70}{5}\right)=0.817 \quad \Rightarrow \quad -\phi\left(\frac{a-70}{5}\right)=0.817-1$$

$$-\phi\left(\frac{a-70}{5}\right)=-0.183 \quad \Rightarrow \quad \phi\left(\frac{a-70}{5}\right)=0.183$$

$$\frac{a-70}{5}=\phi^{-1}(0.813)$$

$$\frac{a-70}{5}=-0.9040 \quad \Rightarrow \quad a-70=5(-0.9040)$$

$$a-70=-4.52 \quad \Rightarrow \quad a=-4.52+70$$

$$a=65.48$$

(iii) Let "*a*" and "*b*" are the required points, then

$$P(a<X<b)=70\%$$

$$P(a<X<b)=0.70$$

As we know that

Total Probability = 1

$$P(X<a)+P(a<X<b)+P(X>b)=1$$

$$P(X<a)+0.70+P(X>b)=1$$

$$P(X<a)+P(X>b)=1-0.70$$

$$P(X<a)+P(X>b)=0.30$$

Then $P(X<a)=P(X>b)=\dfrac{0.30}{2}=0.15$

$$P(X<a)=0.15$$

$$P\left(\frac{X-\mu}{\sigma}<\frac{a-70}{5}\right)=0.15$$

$$P\left(Z<\frac{a-70}{5}\right)=0.15$$

$$P(X>b)=0.15$$

$$P\left(\frac{X-\mu}{\sigma}>\frac{b-70}{5}\right)=0.15$$

$$P\left(Z>\frac{b-70}{5}\right)=0.15$$

$$\phi\left(\frac{a-70}{5}\right)=0.15$$
$$\frac{a-70}{5}=\phi^{-1}(0.15)$$
$$\frac{a-70}{5}=-1.0364 \Rightarrow a-70=5(-1.0364)$$
$$a-70=-5.182 \Rightarrow a=70-5.1825$$
$$a=64.818$$

$$1-\phi\left(\frac{b-70}{5}\right)=0.15$$
$$-\phi\left(\frac{b-70}{5}\right)=0.15-1$$
$$-\phi\left(\frac{b-70}{5}\right)=-0.85 \Rightarrow \quad \phi\left(\frac{b-70}{5}\right)=0.85$$
$$\frac{b-70}{5}=\phi^{-1}(0.85)$$
$$\frac{b-70}{5}=1.0364 \Rightarrow b-70=5(1.0364)$$
$$b-70=55.182 \Rightarrow b=70+5.1825$$
$$b=75.1825$$

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Example# 37: In a normal distribution $\mu=30$ and $\sigma=5$. Find;

(i) A point that has 15% area below it.
(ii) A point that has 28% area above it.
(iii) Two points containing middle 95% area. **D.Y.S.**

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Example# 38: The time required by a nurse to inject a shot of penicillin has been observed to be normally distributed with a mean of $\mu=30$ seconds and a standard deviation of $\sigma=10$ seconds. Find the **(i)** 10th percentile *i.e.* P_{10} **(ii)** 90th percentile *i.e.* P_{90}

***Solution*:** Given $\mu=30$ *and* $\sigma=10$

(i) P_{10} means 10% area below it.

i.e. $P(X<P_{10})=10\%$

$$P(X<P_{10})=0.10$$
$$P\left(\frac{X-\mu}{\sigma}<\frac{P_{10}-30}{10}\right)=0.10$$
$$P\left(Z<\frac{P_{10}-30}{10}\right)=0.10$$
$$\phi\left(\frac{P_{10}-30}{10}\right)=0.10$$
$$\frac{P_{10}-30}{10}=\phi^{-1}(0.10)$$
$$\frac{P_{10}-30}{10}=-1.2816 \quad \Rightarrow \quad P_{10}-30=10(-1.2816)$$
$$P_{10}-30=-12.816 \quad \Rightarrow \quad P_{10}=30-12.816$$
$$P_{10}=17.18\ Seconds$$

(ii) P_{90} means 90% area below it.

i.e. $P(X<P_{90})=90\%$

$$P(X < P_{90}) = 0.90$$

$$P\left(\frac{X-\mu}{\sigma} < \frac{P_{90}-30}{10}\right) = 0.90$$

$$P\left(Z < \frac{P_{90}-30}{10}\right) = 0.90$$

$$\phi\left(\frac{P_{90}-30}{10}\right) = 0.90$$

$$\frac{P_{90}-30}{10} = \phi^{-1}(0.90)$$

$$\frac{P_{90}-30}{10} = 1.2816 \qquad \Rightarrow \qquad P_{90}-30 = 10(1.2816)$$

$$P_{90}-30 = 12.816 \qquad \Rightarrow \qquad P_{90} = 30+12.816$$

$$P_{10} = 42.816\, Seconds$$

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Example# 39: The average life of a certain type of small motors is 10 years with standard deviation of 2 years. The manufacturer replaces free all motors that fail while under guarantee. If he is willing to replace only 3% of the motors that fail. How long a guarantee should he offer? Assume that the lives of the motors follow normal distribution.

***Solution*:** Given $\mu = 10 \ years \quad and \quad \sigma = 2 \ years$

Let G = Guarantee of motors that should be offer by manufacturer

$$P(X < G) = 3\%$$

$$P(X < G) = 0.03$$

$$P\left(\frac{X-\mu}{\sigma} < \frac{G-10}{2}\right) = 0.03$$

$$P\left(Z < \frac{G-10}{2}\right) = 0.03$$

$$\phi\left(\frac{G-10}{2}\right) = 0.03$$

$$\frac{G-10}{2} = \phi^{-1}(0.03)$$

$$\frac{G-10}{2} = -1.8808 \qquad \Rightarrow \qquad G-10 = 2(-1.8808)$$

$$G-10 = -3.7616 \qquad \Rightarrow \qquad P_{10} = 10-3.7616$$

$$G = 6.24\, years$$

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STANDARD NORMAL CUMULATIVE DISTRIBUTION FUNCTION

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-3	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-4	.00003	.00003	.00003	.00003	.00003	.00003	.00002	.00002	.00002	.00002

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Table 10

Quantiles of the Standard Normal Distribution

(Inverse Standard Normal Cumulative Distribution Function)

Tabulated Values: $z_p = \Phi^{-1}(p)$

$$\Phi(z_p) = P(Z \le z_p) = \int_{-\infty}^{z_p} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = p$$

p	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.00	$-\infty$	-3.0902	-2.8782	-2.7478	-2.6521	-2.5758	-2.5121	-2.4573	-2.4089	-2.3656
0.01	-2.3263	-2.2904	-2.2571	-2.2262	-2.1973	-2.1701	-2.1444	-2.1201	-2.0969	-2.0748
0.02	-2.0537	-2.0335	-2.0141	-1.9954	-1.9774	-1.9600	-1.9431	-1.9268	-1.9110	-1.8957
0.03	-1.8808	-1.8663	-1.8522	-1.8384	-1.8250	-1.8119	-1.7991	-1.7866	-1.7744	-1.7624
0.04	-1.7507	-1.7392	-1.7279	-1.7169	-1.7060	-1.6954	-1.6849	-1.6747	-1.6646	-1.6546
0.05	-1.6449	-1.6352	-1.6258	-1.6164	-1.6072	-1.5982	-1.5893	-1.5805	-1.5718	-1.5632
0.06	-1.5548	-1.5464	-1.5382	-1.5301	-1.5220	-1.5141	-1.5063	-1.4985	-1.4909	-1.4833
0.07	-1.4758	-1.4684	-1.4611	-1.4538	-1.4466	-1.4395	-1.4325	-1.4255	-1.4187	-1.4118
0.08	-1.4051	-1.3984	-1.3917	-1.3852	-1.3787	-1.3722	-1.3658	-1.3595	-1.3532	-1.3469
0.09	-1.3408	-1.3346	-1.3285	-1.3225	-1.3165	-1.3106	-1.3047	-1.2988	-1.2930	-1.2873
0.10	-1.2816	-1.2759	-1.2702	-1.2646	-1.2591	-1.2536	-1.2481	-1.2426	-1.2372	-1.2319
0.11	-1.2265	-1.2212	-1.2160	-1.2107	-1.2055	-1.2004	-1.1952	-1.1901	-1.1850	-1.1800
0.12	-1.1750	-1.1700	-1.1650	-1.1601	-1.1552	-1.1503	-1.1455	-1.1407	-1.1359	-1.1311
0.13	-1.1264	-1.1217	-1.1170	-1.1123	-1.1077	-1.1031	-1.0985	-1.0939	-1.0893	-1.0848
0.14	-1.0803	-1.0758	-1.0714	-1.0669	-1.0625	-1.0581	-1.0537	-1.0494	-1.0451	-1.0407
0.15	-1.0364	-1.0322	-1.0279	-1.0237	-1.0194	-1.0152	-1.0110	-1.0069	-1.0027	-0.9986
0.16	-0.9945	-0.9904	-0.9863	-0.9822	-0.9782	-0.9741	-0.9701	-0.9661	-0.9621	-0.9581
0.17	-0.9542	-0.9502	-0.9463	-0.9424	-0.9385	-0.9346	-0.9307	-0.9269	-0.9230	-0.9192
0.18	-0.9154	-0.9116	-0.9078	-0.9040	-0.9002	-0.8965	-0.8927	-0.8890	-0.8853	-0.8816
0.19	-0.8779	-0.8742	-0.8706	-0.8669	-0.8632	-0.8596	-0.8560	-0.8524	-0.8488	-0.8452
0.20	-0.8416	-0.8381	-0.8345	-0.8310	-0.8274	-0.8239	-0.8204	-0.8169	-0.8134	-0.8099
0.21	-0.8064	-0.8030	-0.7995	-0.7961	-0.7926	-0.7892	-0.7858	-0.7824	-0.7790	-0.7756
0.22	-0.7722	-0.7688	-0.7655	-0.7621	-0.7588	-0.7554	-0.7521	-0.7488	-0.7454	-0.7421
0.23	-0.7388	-0.7356	-0.7323	-0.7290	-0.7257	-0.7225	-0.7192	-0.7160	-0.7128	-0.7095
0.24	-0.7063	-0.7031	-0.6999	-0.6967	-0.6935	-0.6903	-0.6871	-0.6840	-0.6808	-0.6776
0.25	-0.6745	-0.6713	-0.6682	-0.6651	-0.6620	-0.6588	-0.6557	-0.6526	-0.6495	-0.6464
0.26	-0.6433	-0.6403	-0.6372	-0.6341	-0.6311	-0.6280	-0.6250	-0.6219	-0.6189	-0.6158
0.27	-0.6128	-0.6098	-0.6068	-0.6038	-0.6008	-0.5978	-0.5948	-0.5918	-0.5888	-0.5858
0.28	-0.5828	-0.5799	-0.5769	-0.5740	-0.5710	-0.5681	-0.5651	-0.5622	-0.5592	-0.5563
0.29	-0.5534	-0.5505	-0.5476	-0.5446	-0.5417	-0.5388	-0.5359	-0.5330	-0.5302	-0.5273
0.30	-0.5244	-0.5215	-0.5187	-0.5158	-0.5129	-0.5101	-0.5072	-0.5044	-0.5015	-0.4987
0.31	-0.4958	-0.4930	-0.4902	-0.4874	-0.4845	-0.4817	-0.4789	-0.4761	-0.4733	-0.4705
0.32	-0.4677	-0.4649	-0.4621	-0.4593	-0.4565	-0.4538	-0.4510	-0.4482	-0.4454	-0.4427
0.33	-0.4399	-0.4372	-0.4344	-0.4316	-0.4289	-0.4261	-0.4234	-0.4207	-0.4179	-0.4152
0.34	-0.4125	-0.4097	-0.4070	-0.4043	-0.4016	-0.3989	-0.3961	-0.3934	-0.3907	-0.3880

(Continued)

p	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.35	-0.3853	-0.3826	-0.3799	-0.3772	-0.3745	-0.3719	-0.3692	-0.3665	-0.3638	-0.3611
0.36	-0.3585	-0.3558	-0.3531	-0.3505	-0.3478	-0.3451	-0.3425	-0.3398	-0.3372	-0.3345
0.37	-0.3319	-0.3292	-0.3266	-0.3239	-0.3213	-0.3186	-0.3160	-0.3134	-0.3107	-0.3081
0.38	-0.3055	-0.3029	-0.3002	-0.2976	-0.2950	-0.2924	-0.2898	-0.2871	-0.2845	-0.2819
0.39	-0.2793	-0.2767	-0.2741	-0.2715	-0.2689	-0.2663	-0.2637	-0.2611	-0.2585	-0.2559
0.40	-0.2533	-0.2508	-0.2482	-0.2456	-0.2430	-0.2404	-0.2378	-0.2353	-0.2327	-0.2301
0.41	-0.2275	-0.2250	-0.2224	-0.2198	-0.2173	-0.2147	-0.2121	-0.2096	-0.2070	-0.2045
0.42	-0.2019	-0.1993	-0.1968	-0.1942	-0.1917	-0.1891	-0.1866	-0.1840	-0.1815	-0.1789
0.43	-0.1764	-0.1738	-0.1713	-0.1687	-0.1662	-0.1637	-0.1611	-0.1586	-0.1560	-0.1535
0.44	-0.1510	-0.1484	-0.1459	-0.1434	-0.1408	-0.1383	-0.1358	-0.1332	-0.1307	-0.1282
0.45	-0.1257	-0.1231	-0.1206	-0.1181	-0.1156	-0.1130	-0.1105	-0.1080	-0.1055	-0.1030
0.46	-0.1004	-0.0979	-0.0954	-0.0929	-0.0904	-0.0878	-0.0853	-0.0828	-0.0803	-0.0778
0.47	-0.0753	-0.0728	-0.0702	-0.0677	-0.0652	-0.0627	-0.0602	-0.0577	-0.0552	-0.0527
0.48	-0.0502	-0.0476	-0.0451	-0.0426	-0.0401	-0.0376	-0.0351	-0.0326	-0.0301	-0.0276
0.49	-0.0251	-0.0226	-0.0201	-0.0175	-0.0150	-0.0125	-0.0100	-0.0075	-0.0050	-0.0025
0.50	0.0000	0.0025	0.0050	0.0075	0.0100	0.0125	0.0150	0.0175	0.0201	0.0226
0.51	0.0251	0.0276	0.0301	0.0326	0.0351	0.0376	0.0401	0.0426	0.0451	0.0476
0.52	0.0502	0.0527	0.0552	0.0577	0.0602	0.0627	0.0652	0.0677	0.0702	0.0728
0.53	0.0753	0.0778	0.0803	0.0828	0.0853	0.0878	0.0904	0.0929	0.0954	0.0979
0.54	0.1004	0.1030	0.1055	0.1080	0.1105	0.1130	0.1156	0.1181	0.1206	0.1231
0.55	0.1257	0.1282	0.1307	0.1332	0.1358	0.1383	0.1408	0.1434	0.1459	0.1484
0.56	0.1510	0.1535	0.1560	0.1586	0.1611	0.1637	0.1662	0.1687	0.1713	0.1738
0.57	0.1764	0.1789	0.1815	0.1840	0.1866	0.1891	0.1917	0.1942	0.1968	0.1993
0.58	0.2019	0.2045	0.2070	0.2096	0.2121	0.2147	0.2173	0.2198	0.2224	0.2250
0.59	0.2275	0.2301	0.2327	0.2353	0.2378	0.2404	0.2430	0.2456	0.2482	0.2508
0.60	0.2533	0.2559	0.2585	0.2611	0.2637	0.2663	0.2689	0.2715	0.2741	0.2767
0.61	0.2793	0.2819	0.2845	0.2871	0.2898	0.2924	0.2950	0.2976	0.3002	0.3029
0.62	0.3055	0.3081	0.3107	0.3134	0.3160	0.3186	0.3213	0.3239	0.3266	0.3292
0.63	0.3319	0.3345	0.3372	0.3398	0.3425	0.3451	0.3478	0.3505	0.3531	0.3558
0.64	0.3585	0.3611	0.3638	0.3665	0.3692	0.3719	0.3745	0.3772	0.3799	0.3826
0.65	0.3853	0.3880	0.3907	0.3934	0.3961	0.3989	0.4016	0.4043	0.4070	0.4097
0.66	0.4125	0.4152	0.4179	0.4207	0.4234	0.4261	0.4289	0.4316	0.4344	0.4372
0.67	0.4399	0.4427	0.4454	0.4482	0.4510	0.4538	0.4565	0.4593	0.4621	0.4649
0.68	0.4677	0.4705	0.4733	0.4761	0.4789	0.4817	0.4845	0.4874	0.4902	0.4930
0.69	0.4958	0.4987	0.5015	0.5044	0.5072	0.5101	0.5129	0.5158	0.5187	0.5215
0.70	0.5244	0.5273	0.5302	0.5330	0.5359	0.5388	0.5417	0.5446	0.5476	0.5505
0.71	0.5534	0.5563	0.5592	0.5622	0.5651	0.5681	0.5710	0.5740	0.5769	0.5799
0.72	0.5828	0.5858	0.5888	0.5918	0.5948	0.5978	0.6008	0.6038	0.6068	0.6098
0.73	0.6128	0.6158	0.6189	0.6219	0.6250	0.6280	0.6311	0.6341	0.6372	0.6403
0.74	0.6433	0.6464	0.6495	0.6526	0.6557	0.6588	0.6620	0.6651	0.6682	0.6713

p	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.75	0.6745	0.6776	0.6808	0.6840	0.6871	0.6903	0.6935	0.6967	0.6999	0.7031
0.76	0.7063	0.7095	0.7128	0.7160	0.7192	0.7225	0.7257	0.7290	0.7323	0.7356
0.77	0.7388	0.7421	0.7454	0.7488	0.7521	0.7554	0.7588	0.7621	0.7655	0.7688
0.78	0.7722	0.7756	0.7790	0.7824	0.7858	0.7892	0.7926	0.7961	0.7995	0.8030
0.79	0.8064	0.8099	0.8134	0.8169	0.8204	0.8239	0.8274	0.8310	0.8345	0.8381
0.80	0.8416	0.8452	0.8488	0.8524	0.8560	0.8596	0.8632	0.8669	0.8706	0.8742
0.81	0.8779	0.8816	0.8853	0.8890	0.8927	0.8965	0.9002	0.9040	0.9078	0.9116
0.82	0.9154	0.9192	0.9230	0.9269	0.9307	0.9346	0.9385	0.9424	0.9463	0.9502
0.83	0.9542	0.9581	0.9621	0.9661	0.9701	0.9741	0.9782	0.9822	0.9863	0.9904
0.84	0.9945	0.9986	1.0027	1.0069	1.0110	1.0152	1.0194	1.0237	1.0279	1.0322
0.85	1.0364	1.0407	1.0451	1.0494	1.0537	1.0581	1.0625	1.0669	1.0714	1.0758
0.86	1.0803	1.0848	1.0893	1.0939	1.0985	1.1031	1.1077	1.1123	1.1170	1.1217
0.87	1.1264	1.1311	1.1359	1.1407	1.1455	1.1503	1.1552	1.1601	1.1650	1.1700
0.88	1.1750	1.1800	1.1850	1.1901	1.1952	1.2004	1.2055	1.2107	1.2160	1.2212
0.89	1.2265	1.2319	1.2372	1.2426	1.2481	1.2536	1.2591	1.2646	1.2702	1.2759
0.90	1.2816	1.2873	1.2930	1.2988	1.3047	1.3106	1.3165	1.3225	1.3285	1.3346
0.91	1.3408	1.3469	1.3532	1.3595	1.3658	1.3722	1.3787	1.3852	1.3917	1.3984
0.92	1.4051	1.4118	1.4187	1.4255	1.4325	1.4395	1.4466	1.4538	1.4611	1.4684
0.93	1.4758	1.4833	1.4909	1.4985	1.5063	1.5141	1.5220	1.5301	1.5382	1.5464
0.94	1.5548	1.5632	1.5718	1.5805	1.5893	1.5982	1.6072	1.6164	1.6258	1.6352
0.95	1.6449	1.6546	1.6646	1.6747	1.6849	1.6954	1.7060	1.7169	1.7279	1.7392
0.96	1.7507	1.7624	1.7744	1.7866	1.7991	1.8119	1.8250	1.8384	1.8522	1.8663
0.97	1.8808	1.8957	1.9110	1.9268	1.9431	1.9600	1.9774	1.9954	2.0141	2.0335
0.98	2.0537	2.0748	2.0969	2.1201	2.1444	2.1701	2.1973	2.2262	2.2571	2.2904
0.99	2.3263	2.3656	2.4089	2.4573	2.5121	2.5758	2.6521	2.7478	2.8782	3.0902
1.00	∞									